

PERSONAL FINANCE TRACKER

Python Project Report

Group Members

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1. Abstract & Introduction

Managing personal finances is essential for maintaining financial stability and making informed decisions. However, many individuals find it difficult to track their daily expenses manually, which often leads to poor budgeting and lack of financial awareness. To address this problem, a Personal Finance Tracker application has been developed using Python.

This application provides a simple and user-friendly interface to record, store, and analyze daily expenses. It integrates multiple technologies such as Tkinter for the graphical user interface, Pandas for data handling, NumPy for numerical computations, Matplotlib for data visualization, and MongoDB for database management.

The system allows users to perform essential operations such as adding new transactions, viewing stored data, deleting records, and analyzing spending patterns. It also provides graphical insights through bar charts and pie charts, helping users understand their expenditure distribution.

Overall, this project demonstrates how basic data science tools and database integration can be combined to build an effective and practical financial management system.

2. Problem Statement & Design

Problem Statement

People often struggle to track their expenses manually, which leads to poor financial management. There is a need for a simple system that allows users to:

- Record daily expenses
- View and manage data
- Analyze spending patterns

System Design

The application follows a simple modular design:

- **Frontend (GUI): Tkinter**
- **Backend Logic: Python functions**
- **Database: MongoDB**
- **Data Processing: Pandas & NumPy**
- **Visualization: Matplotlib**

Workflow

- 1. Load dataset from CSV**
 - 2. Store data in MongoDB**
 - 3. Perform CRUD operations**
 - 4. Analyze and visualize data**
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3. Methodology

Step 1: Data Collection

- Dataset contains fields: Date, Category, Amount
- Loaded using Pandas

Step 2: Data Cleaning

- Converted Date to datetime format
- Converted Amount to numeric
- Removed invalid values

Step 3: Database Integration

- MongoDB used for storing transactions
- Inserted dataset into database
- Enabled CRUD operations

Step 4: GUI Development

- Tkinter used to create input fields and buttons
- Listbox used to display records

Step 5: Functional Implementation

- Add transaction (Create)
- View data (Read)

- Delete selected/all (Delete)
- Analyze data (Statistics + Graphs)

Step 6: Data Analysis

- Total spending using NumPy
- Average, minimum, maximum calculation
- Category-wise grouping using Pandas

Step 7: Visualization

- Bar chart → category spending
- Pie chart → distribution of expenses

4. Tools Used

Tool	Purpose
Python	Programming language
Tkinter	GUI development
MongoDB	Database storage
Pandas	Data handling
NumPy	Numerical operations
Matplotlib	Data visualization
PyMongo	MongoDB connection

5. Results

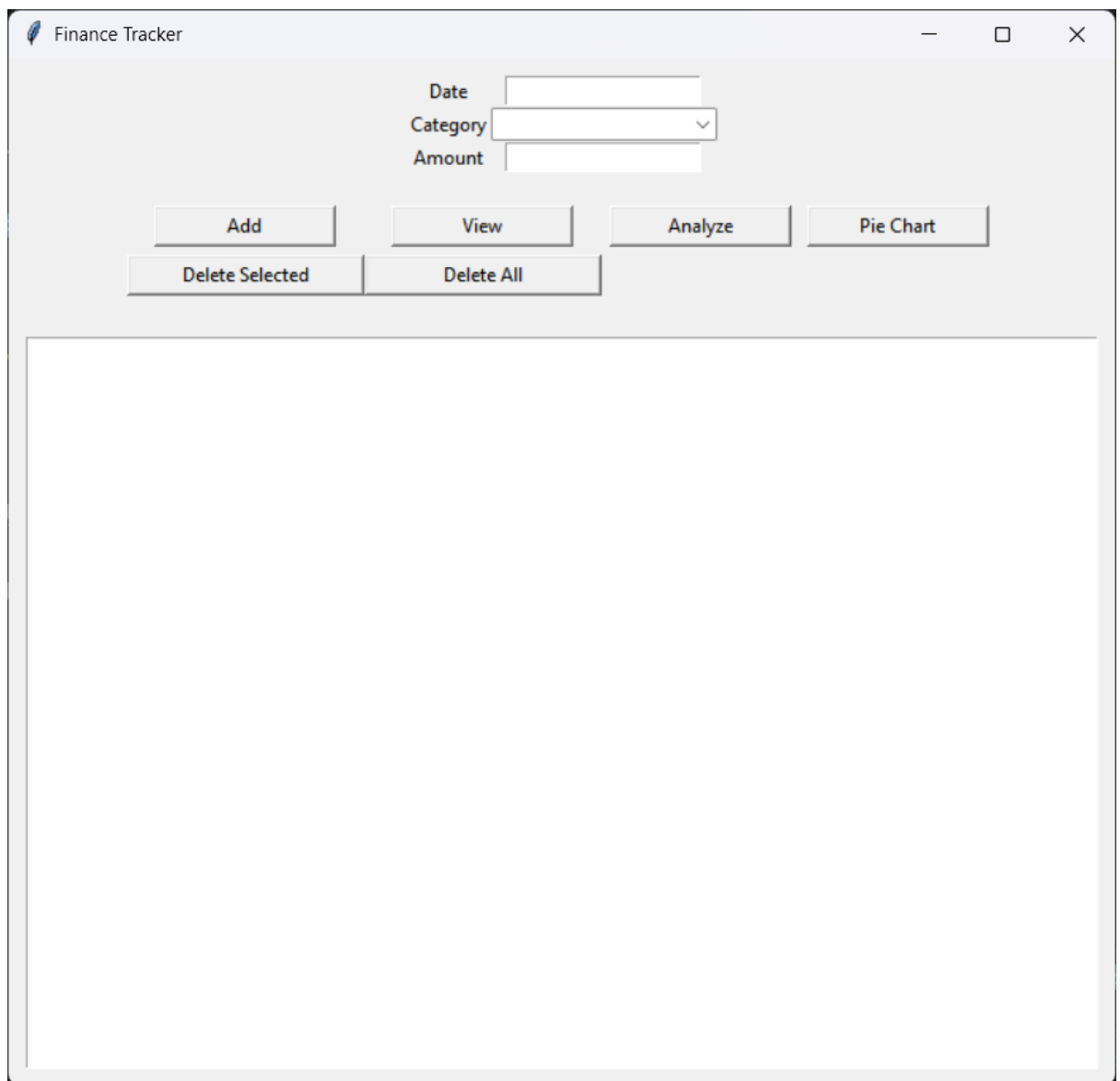
The application successfully performs:

- Adding new transactions
- Viewing stored data
- Deleting selected and all records
- Displaying statistics:
 - Total spending
 - Average spending

- Minimum spending
 - Maximum spending
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Screenshots

1. GUI Interface



The screenshot displays the 'Finance Tracker' application window. The title bar at the top shows the application name and standard window controls (minimize, maximize, close). The main interface is divided into two sections. The top section contains input fields for 'Date', 'Category' (a dropdown menu), and 'Amount'. Below these inputs is a row of five buttons: 'Add', 'View', 'Analyze', 'Pie Chart', and 'Delete Selected'. The 'Delete All' button is positioned below 'Delete Selected'. The bottom section of the window is a large, empty rectangular area, likely intended for displaying a list of transactions or a chart.

2. Table View (Stored Data)

Finance Tracker

Date

Category

Amount

Add

View

Analyze

Pie Chart

Delete Selected

Delete All

2024-01-01 00:00:00	Transport	3279
2024-01-02 00:00:00	Entertainment	2080
2024-01-05 00:00:00	Food	13249
2024-01-05 00:00:00	Entertainment	14800
2024-01-08 00:00:00	Shopping	10856
2024-01-09 00:00:00	Shopping	3630
2024-01-09 00:00:00	Rent	9227
2024-01-10 00:00:00	Food	2765
2024-01-12 00:00:00	Rent	7214
2024-02-02 00:00:00	Entertainment	999
2024-02-02 00:00:00	Shopping	4098
2024-02-04 00:00:00	Shopping	9647
2024-02-04 00:00:00	Health	4214
2024-02-05 00:00:00	Entertainment	14836
2024-02-05 00:00:00	Entertainment	11344
2024-02-06 00:00:00	Utilities	2820
2024-02-06 00:00:00	Rent	6336
2024-02-06 00:00:00	Entertainment	6328
2024-02-07 00:00:00	Health	13261
2024-02-08 00:00:00	Food	5799
2024-02-08 00:00:00	Health	13181
2024-02-08 00:00:00	Food	10833
2024-02-10 00:00:00	Shopping	2528
2024-02-10 00:00:00	Shopping	12813
2024-02-10 00:00:00	Health	4411
2024-02-11 00:00:00	Entertainment	6994
2024-02-12 00:00:00	Food	10924
2024-03-02 00:00:00	Shopping	11656
2024-03-03 00:00:00	Utilities	12743
2024-03-05 00:00:00	Rent	2239
2024-03-09 00:00:00	Utilities	12711
2024-03-09 00:00:00	Shopping	12215
2024-03-10 00:00:00	Utilities	109
2024-03-12 00:00:00	Food	10333
2024-04-01 00:00:00	Rent	1740
2024-04-01 00:00:00	Food	8421
2024-04-03 00:00:00	Food	6627
2024-04-05 00:00:00	Entertainment	4393

3. Analysis Output

Analysis

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Total = 3232492

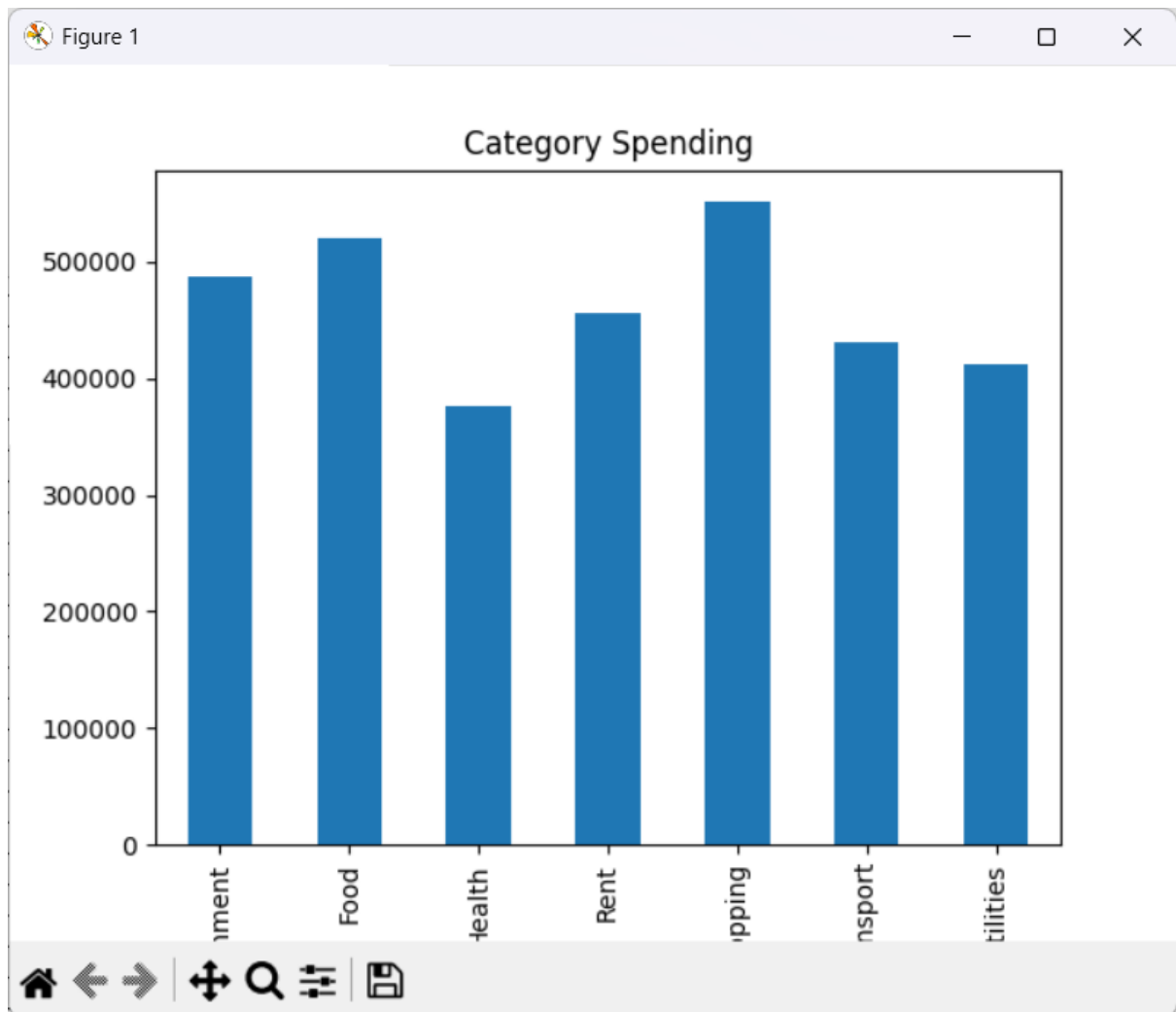
Average = 7942.240786240786

Min Spend = 101

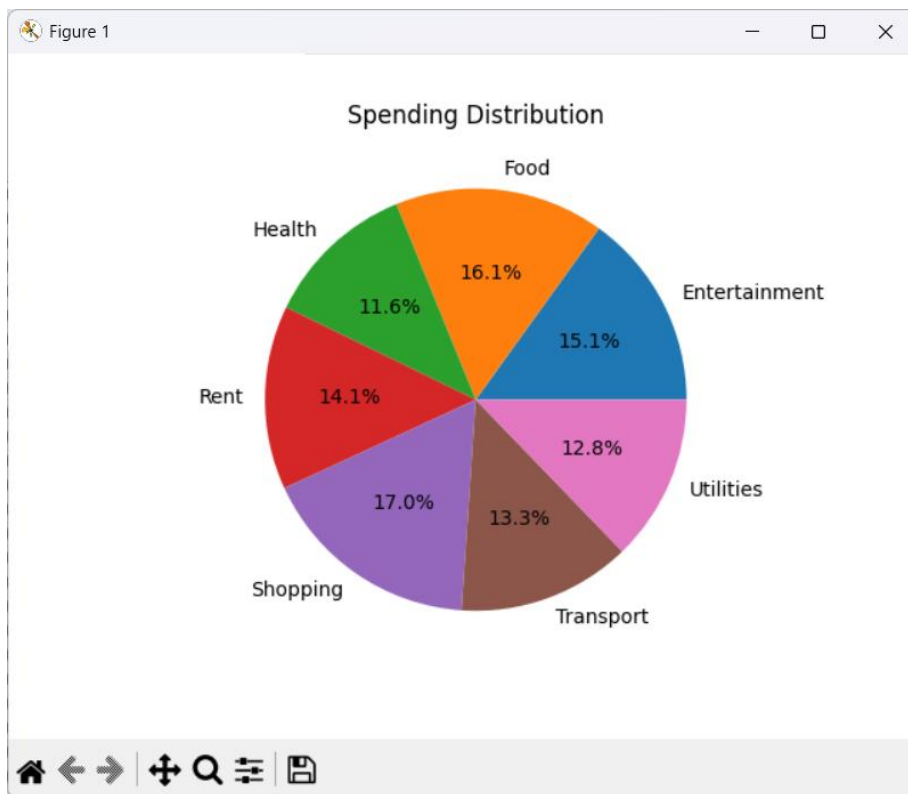
Max Spend = 14999

OK

4.Graph Visualization



5. Pie chart



7. Conclusion & Future Scope

Conclusion

The Personal Finance Tracker successfully demonstrates the integration of GUI, database, and data analysis tools in Python. It provides a simple and effective way to manage and analyze personal expenses.

Future Scope

The project can be further improved by:

- Adding update/edit functionality
 - Implementing search and filter options
 - Creating monthly/yearly reports
 - Using advanced GUI frameworks (like PyQt)
 - Adding login/authentication system
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